



Carotenoids... Or Is It “Carrot - Nerds”?

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It's Personal

Carotenoids... How did I get on this topic? Well, let me tell you. I was on the phone several nights ago with a dear friend in Dallas Texas, and as usual we were comparing notes on the latest eye health information we found. You see we are both over 60 years young, and ‘seeing’ age-related eye issues encroaching on our lives. We are trying to approach healing (or at least slowing the progression of our abnormalities) in a naturopathic way... especially when we find it difficult locating an optometrist or ophthalmologist that practices functional medicine or will at least consider functional nutrition as an option or supplemental method of healing. Anyway, she mentioned that her eye doctor suggested she start taking a ‘*Carotenoid Complex*’. I was curious as to what was in a ‘*Carotenoid Complex*’ and how it differed from what I am currently taking as eye health supplements and the diet I am following. So, here is what I found...

What are Carotenoids?

Carotenoids are typically considered the plant pigments that give certain flowers, fruits and vegetables the red, orange and yellow hues. While this is true, carotenoids also provide the striking colours found in many of our animals and in microorganisms, as we shall see shortly. According to an article in “*Methods in Enzymology*”: “*They [Carotenoids] have other, less obvious roles that make them essential components in oxygenic photosynthesis. Without carotenoids, photosynthesis and all life in an aerobic atmosphere would be impossible. They have also been implicated in the maintenance of good health in animals, including humans.*” (1)

Did you know? Flamingos have pink feathers because of what they eat. Yes, It's true! It is part of their biological makeup. These beautiful boldly coloured birds are pink because they primarily consume blue-green algae (cyanobacteria), especially *Spirulina platensis*, along with various diatoms, red algae, and small crustaceans like brine shrimp. The microscopic red algae and the brine shrimp are rich in carotenoid pigments—specifically canthaxanthin and beta-carotene. The high levels of carotenoid pigments (specifically alpha and beta carotenoids) in the flamingos, when digested and metabolized, turn their feathers and skin a brilliant pink, red or orange! Now that definitely fits the saying “*You are what you eat! And what your body can do with what you eat!*”

Carotenoids, a major food micronutrient found in a nutrition-rich diet, are fat-soluble compounds that should be consumed with some ‘healthy’ fats, such as olive oil, nuts, or avocado. Their primary benefits are that they are anti-inflammatory and anti-oxidants. They help to battle the free radicals and reduce the risk of chronic disease. There are more than 600 types of carotenoids (2) which include:

- **alpha– and beta-carotene:** As needed the body converts these into Vitamin A for a healthy immune function and vision, as well as, cell growth.



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- **beta-cryptoxanthin:** A powerful anti-oxidant and provitamin A carotenoid that act as a precursor to Vitamin A. As a provitamin beta-cryptoxanthin, is converted to Vitamin A in the body, supporting eye, skin and immune health. It also promotes bone health, fostering bone formation and helping osteoporosis prevention.
- **lutein and zeaxanthin:** Found primarily in the lens and retina of the eye, they are responsible for protecting the eye against age-related abnormalities, such as age-related macular degeneration (AMD) and cataracts. (2) They also help protect the eye from harmful blue light. “By lowering the amount of blue light that’s able to enter the eye, carotenoids help preserve healthy cells and slow down oxidative damage that leads to vision loss.” (2)
- **lycopene:** Tomatoes are the best-known source of this potent antioxidant, especially when cooked. Lycopene is associated with a reduced risk of certain cancers (especially prostate cancer), (2) hypertension and cardiovascular disease, as it helps protect lipoproteins from oxidation. (3)
- **astaxanthin:** The carotenoid is known for the natural red-orange colouring found in foods like salmon, krill, and lobster. Recognized for its strong antioxidant capabilities, astaxanthin supports the immune system and has shown benefits for skin health, including improved elasticity and reduced wrinkles. “Once converted, vitamin A helps prevent keratinization of the skin, which occurs when the epithelial cells lose their moisture and become dry. Active vitamin A fights skin cancer because it helps stop oxidative damage and control malignant cells in the body, while facilitating with the process of correct cell differentiation.” (2)

Provitamin A Carotenoids Vice Vitamin A (Retinoids) – What’s the Difference? Good question... Let’s take a quick look! Vitamin A, also known as **preformed Vitamin A** as it is the active form which the body can use without conversion. It is found primarily in animal-based foods such as eggs, dairy products, fatty fish and liver. On the other hand, **Provitamin A carotenoids** are found in the vibrant fruits and vegetables, as previously mentioned. When a food containing carotenoids is consumed, the body converts certain carotenoids, such as beta-carotene to active Vitamin A as needed by the body. (4)

Am I Having a ‘Carotenoid Complex’? So, next question is... What is a ‘carotenoid complex’? I am glad you asked. It is a supplement (*Another pill... Yikes! And yes, I went to a few websites to read labels. Do you really need to take a pill when there are so many vibrant fruits and vegetables packed with carotenoids? Well, that is up to you and your doctor to decide. I am just providing information.*). Back to Carotenoid Complexes... They are a blend of plant-derived pigments that act as antioxidants, supporting **immune function, vision, heart health, and skin protection**. These carotenoid complexes typically offer a broad spectrum of carotenoids, not just a single type like beta-carotene, to provide comprehensive health benefits. Carotenoids are present in a diet rich in fruits and vegetables. Thus, supplements containing carotenoids are intended either to boost current carotenoid intake in individuals who already eat a carotenoid-rich diet, or to provide carotenoids to those whose diet contains only low amounts of them.

The Functional Nutrition Approach:

As you can probably can guess, a lot of foods are carotenoid rich and to eat a variety of nutrient-rich foods is best. Registered dietitian Devan Peart, RD, MHSc puts it this way, “Micronutrients, including carotenoids, are widely distributed amongst foods. If you always eat the same vegetables or the same fruits, you’ll get lots of certain nutrients and miss out on others.” (3) Carotenoids are one of the major food micronutrients within the human diet with



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the key carotenoids present in food and human tissues being: alpha-carotene, beta-carotene, beta-cryptoxanthin, lycopene, lutein and zeaxanthin.

Getting 'Back to the Basics':

So, let's look at getting back to the functional nutrition basics... and look at the **Food Rainbow!**

What do we mean by "Eat the Rainbow"? When it comes to food, "Eating the Rainbow" in Functional Nutrition means consuming a wide variety of colorful fresh fruits and vegetables daily to ensure a broad intake of essential nutrients, antioxidants, and phytochemicals. (The best daily recommendation is at least 2-3 fruits and/or vegetables at each meal.) The phytonutrients found in plants are natural compounds that give them their, brilliant colour, flavour and healthful benefits. For example, the green fruits and vegetables such as kale, spinach, broccoli and avocado are teaming with vitamins (such as folate), minerals, lutein, chlorophyll, and cancer-fighting anti-oxidants which are detoxifying and support brain, heart, mitochondria function, and the immune system. The reds, such as tomatoes, apples, raspberries, and goji berries are high in lycopene and anthocyanins which support heart health and reduce cancer risk. The orange and yellow fruits and vegetables are packed the **carotenoids** boosting skin, eye and immune health. (3) NOTE: The "Food Rainbow" includes 7 primary colours; however, we are only focusing on the carotenoid-rich hues: red, orange, yellow, and green (ROYG). See Figure 1.

Concluding Remarks

Carotenoids are the pigments found in the various algae, fruits and vegetables. They are a major micronutrient found in a nutrition-dense diet. Of the 600+ carotenoids, beta-carotene, alpha-carotene, beta-cryptoxanthin, lutein, zeaxanthin and lycopene are the most common types. Their primary benefits include: anti-inflammatory and anti-oxidant which help to battle the free radicals, promoting a healthy immune system, and reduce the risk of chronic disease; maintaining eye and vision health by protecting the eyes from harmful blue light; and protecting the skin from environmental damage which improves skin health, including skin elasticity, and reducing fine lines and wrinkles. *And we all want and need that!*

When we look at the "**Food Rainbow**", these mighty micronutrients can be found in the yummy and brightly coloured fruits and vegetables within the red, orange, yellow, and green hues – reasoning that it is why we need to "Eat the Rainbow" daily for optimal nutrition and health! It is best to eat a variety of nutrient-dense fruits and vegetables with each meal. The latest USDA "Dietary Guidelines for Americans" recommends 3 servings of vegetables and 2 servings of fruit per day. (5) Another cool factoid we need to remember is that carotenoids are actually enhanced (i.e. improved availability) by cooking, which is atypical for most foods! (2) **So, On Your Mark! Get Set! Go! Go "Eat the Rainbow"!** and go get those mighty carotenoids and other vitamins and minerals your body needs! And one final note... **Enjoy a Healthier You!!!**



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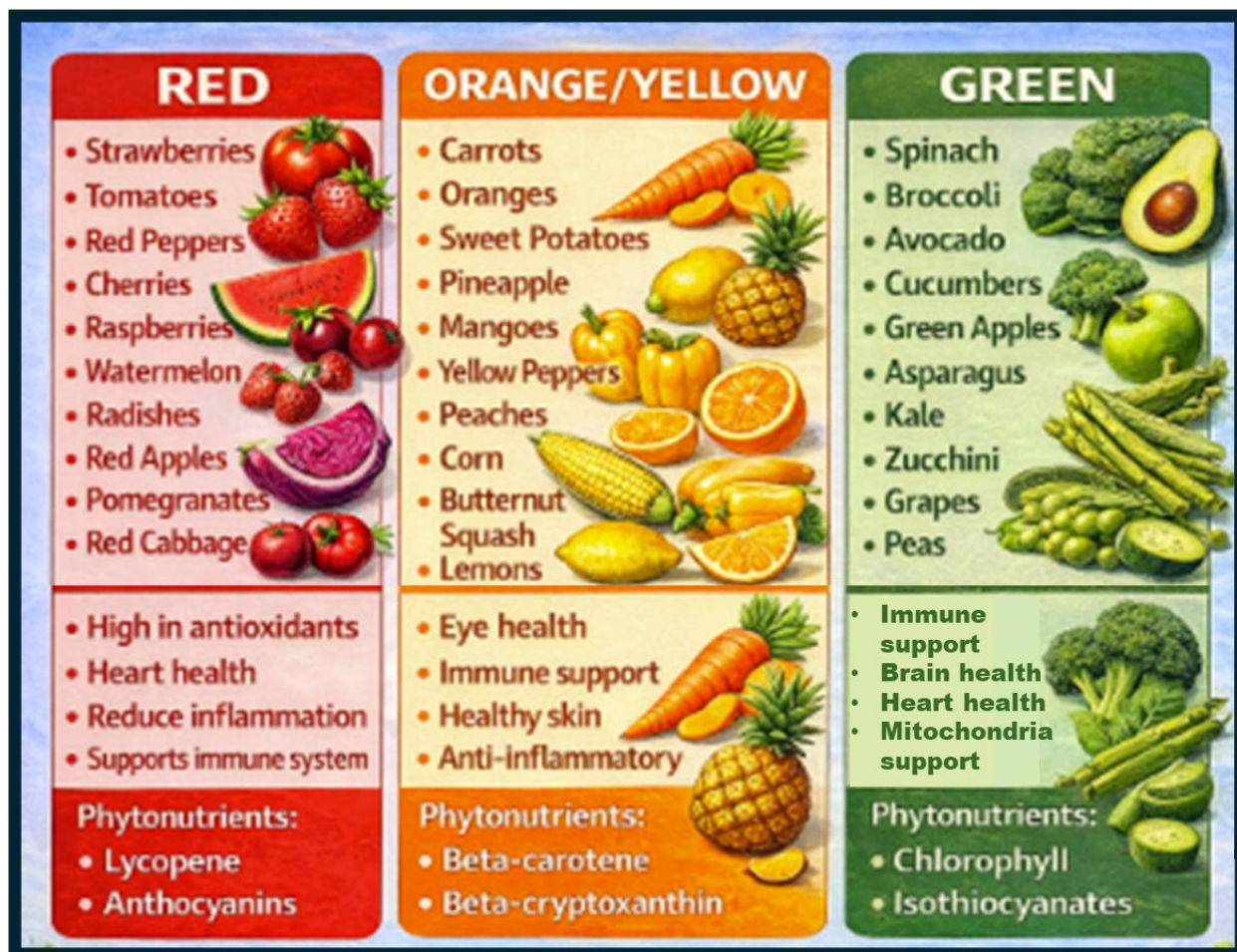


Figure 1. "Food Rainbow" Showing ROYG

- Image created using ChatGPT

Reference material available upon request.

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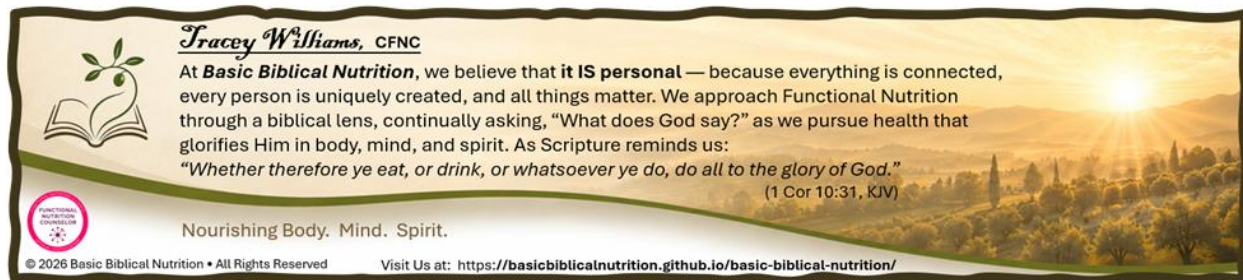
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Tracey Williams is not a medical doctor. She is a Certified Functional Nutrition Counselor/Researcher. Tracey Williams does not diagnose or treat any medical illness or condition.



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At **Basic Biblical Nutrition**, we believe that **it IS personal** — because everything is connected, every person is uniquely created, and all things matter. We approach Functional Nutrition through a biblical lens, continually asking, “What does God say?” as we pursue health that glorifies Him in body, mind, and spirit. As Scripture reminds us:

“Whether therefore ye eat, or drink, or whatsoever ye do, do all to the glory of God.”
(1 Cor 10:31, KJV)

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